METRIC

MIL-PRF-8565/15A
W/AMENDMENT 1
19 April 2010
SUPERSEDING
MIL-PRF-8565/15A
14 July 2009

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MEDIUM-RATE, TYPE 1, MAINTENANCE-FREE, 24-VOLT, 35-AMPERE-HOUR

Reactivated after 14 July 2009 and may be used for new and existing designs and applications.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-8565.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the batteries covered by this specification sheet shall be D8565/15-1.

2. Dimensions and weight. The dimensions of the battery shall be as shown on figure 1. The weight of the battery shall be not greater than 40.82 kilograms (90.0 pounds) (see 3.8.3 and 4.5.3 of MIL-PRF-8565).

3. Capacity. The rated capacity of the battery shall be not less than 35 ampere-hours (Ah), based on its ability to conform to the life test final discharge (35.0 Ah/1 Hr/24°C/20.0 V). See 3.8.6, as modified by MIL-PRF-8565/15, and 4.5.7 and 6.10 of MIL-PRF-8565.

4. Service life. The battery's service life (useful life) in aircraft shall be not less than 4 years.

5. Qualification inspection and conformance inspection. Table I lists the required inspections.

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6. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565 except as follows.

6.1 The following paragraphs are not applicable:

3.5.10 Heaters.
3.6.5 Marking for batteries with heaters.
3.8.16 Output performance.
3.8.17 Cold and hot temperature charge.
4.5.28 Output performance test.
4.5.29 Cold and hot temperature charge test.

6.2 Add the following paragraph, table II and figure 2:

"3.8.18 Vibration. The battery shall maintain a voltage of not less than 20.0 volts during each of the discharges of steps 4.5.19.c and 4.5.19.d, herein. The battery shall also meet the requirements specified in 3.7 and 3.8.10 of MIL-PRF-8565 after being tested in accordance with 4.5.19, herein."

6.3 Modify the following paragraphs:

6.3.1 Change "18.0 volts" to "20.0 volts" in the following paragraphs:

4.5.6 Electrical test preparation, first sentence.
4.5.7 Capacity discharge, second sentence.
4.5.11 Capacity discharge at 49°C (120°F), third sentence.
4.5.12 Charge and discharge test at low temperature, step c and step f, second sentence.
4.5.17 Temperature rise and float test, step e, second sentence.
4.5.22 Ground storage test, step d and i.
4.5.23 Physical integrity at high temperature test, step b and k.
4.5.26 Shelf life test, sixth sentence.
TABLE III Capacity performance requirements, requirements (1), (2), and (3).

6.3.2 Modify the following paragraphs as stated:

3.5.5 Receptacles. Add the following: "For D8565/15-1, the female receptacle (front view) shall be dimensionally equal to MS3509, style 1. The receptacle may be integrally molded in the container."

3.6.4 Battery caution marking. Delete MIL-PRF-8565 paragraph in its entirety and substitute the following: "Marking for Label 2 shown on figure 1 shall show the following with lettering height not less than 6.35mm (0.25 inch). The marking shall be in white or black print.

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MAINTENANCE-FREE BATTERY

DO NOT REMOVE COVER

PROCESS THIS BATTERY IN ACCORDANCE WITH APPROVED PROCEDURES ONLY

SEE NAVAIR 17-15BAD-1 OR APPLICABLE SERVICE T.O."

3.8.6 Capacity and electrical performance. Delete the entire text and insert: “The rated capacity of the battery shall not be less than 35 ampere-hours, based on its ability to conform to the life test final discharge. The battery shall produce not less than 20.0 volts at the C-rate for not less than 60 minutes at 24°C (35.0 Ah/1 Hr/24°C/20.0 V) after being tested in accordance with 4.5.7, as modified by MIL-PRF-8565/15. The discharge requirements and associated parameters shall be as shown in table III (see 6.10). Both table III and 6.10 are in MIL-PRF-8565. Battery capacity shall vary not greater than ± 5 percent within a given inspection lot. The tolerance of the discharge current shall be ± 1 percent of the specified value.

3.8.7 Constant voltage discharge. Delete the first sentence and insert: “The battery shall comply with the following requirements during the 14.0-volt constant voltage discharge:

Requirement	24°C	-26°C
a. Minimum Current at Peak Value (Time = 0+)	1300 amperes	800 amperes
b. Minimum Current after 15 seconds	1000 amperes	450 amperes
c. Minimum Capacity delivered after 60 seconds	13.67 ampere-hours	5.93 ampere-hours

Inability to deliver the specified minimum discharge current and capacity shall constitute a test failure.”

3.8.9 Cycling. In the first sentence, delete "100 cycles" and substitute "135 cycles".

3.8.13.3 Deep discharge recovery. Delete the second sentence and insert: “After such storage, the battery shall deliver not less than 35 ampere-hours after being tested in accordance with 4.5.27 of MIL-PRF-8565.”

4.4.2 Constant potential charging method. Delete MIL-PRF-8565 paragraph in its entirety and substitute the following: "Unless otherwise specified, the battery shall be charged at a constant potential of 28.5 ± 0.1 volts for 4.0 ± 0.1 hours with a power source having a current limit of 25 ± 1 amperes."

4.5.10 Cycling test. In the second sentence, delete "100 cycles" and substitute "135 cycles". In the third sentence, delete "2.0-hour" and substitute "4.0-hour".

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4.5.19 Vibration test. Delete the text and insert: "This test consists of the following steps.

"a. Mount the battery, including electrical connector and cables, to the vibration test apparatus in accordance with 4.4.5 of MIL-PRF-8565.

"b. Constant potential charge each battery in accordance with 4.4.2, herein.

"c. Discharge the battery at a rate of 227.5 amperes for 1.0 minute. Monitor the battery's terminal voltage during discharge. Examine the battery in accordance with 3.8.18, herein.

"d. Charge the battery in accordance with 4.4.2, herein during the vibration exposure. Subject the battery to the random vibration test levels specified in table II and shown on figure 2, herein. The vibration test time for each axis shall be 60.0 +1.0/-0.0 minutes. Conduct the vibration along each of the battery's three mutually perpendicular axes. Before each axis of vibration, discharge the battery at a rate of 227.5 amperes for 1.0 minute. Monitor the battery's terminal voltage during discharge. Examine the battery for compliance with 3.8.18, herein.

"e. Dismount the battery from the vibration test apparatus.

"f. Discharge the battery at a rate of 227.5 amperes for 3.0 minutes. The battery shall maintain a terminal voltage of not less than 14.0 volts.

"g. Charge the battery in accordance with 4.4.2, herein.

"h. Examine the battery per the requirements of 3.8.18, herein."

TABLE I. Qualification inspection and conformance inspection of batteries.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph 1/	Test paragraph 1/	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
1	INCOMING INSPECTION	X	X	X	3.8.1(G)	4.5.1(G)	X	X	X
2	VISUAL AND MECHANICAL EXAMINATION	X	X	X	3.5 (S), 3.6 (S), 3.8.2 (G) table II (G)	4.5.2 (G)	X	X	X
3	DIMENSIONS AND WEIGHT	X	X	X	Reqmt 2 (N), 3.8.3 (G)	4.5.3 (G)	X 2/	X 2/	X

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TABLE I. Qualification inspection and conformance inspection of batteries – Continued.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph 1/	Test paragraph 1/	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
4	DIELECTRIC STRENGTH AND INSULATION RESISTANCE TEST	X	X	X	3.7 (G), 3.8.4 (G)	4.5.4 (G)	X 2/	X	X
5	STRENGTH OF VENT TUBES TEST	X	X	X	3.7 (G), 3.8.5 (G)	4.5.5 (G)	X 2/	X	X
6	ELECTRICAL TEST PREPARATION	X	X	X		4.5.6 (S)		X	X
7	CAPACITY DISCHARGE	X	X	X	Reqmt 3 (N), 3.8.6 (S), Reqmt 1 of table III (S)	4.5.7 (S)	X	X	X
8	CONSTANT VOLTAGE DISCHARGE	X	X		3.7 (G), 3.8.7 (S), Reqmt 4 of table III (G)	4.5.8 (G)	X	X	
9	STRENGTH OF RECEPTACLE TEST			X	3.7 (G), 3.8.8 (G)	4.5.9 (G)			
10	CYCLING TEST	X			Reqmt 4 (N), 3.8.9 (S), Reqmt 1 of table III (S)	4.5.10 (S)		X	
11	CAPACITY DISCHARGE AT 49 °C (120 °F)		X		3.8.6 (S), Reqmt 2 of table III (S)	4.5.11 (S)			
12	CHARGE AND DISCHARGE TEST AT LOW TEMPERATURE		X		3.8.6 (S), Reqmt 3 of table III (S)	4.5.12 (S)			X
13	DISCHARGE WHILE INVERTED		X		3.5.9 (G), Reqmt 5 of table III (G)	4.5.13 (G)			X
14	ALTITUDE TEST		X		3.5.9 (G), 3.7 (G), 3.8.10 (G)	4.5.14 (G)			X
15	SHOCK TEST (BASIC DESIGN)	X			3.7 (G), 3.8.10 (G)	4.5.15 (G)			
16	TEMPERATURE SHOCK TEST	X			3.7 (G), 3.8.10 (G)	4.5.16 (G)		X	
17	TEMPERATURE RISE AND FLOAT TEST		X		3.7 (G), 3.8.11 (G), Reqmt 1 of table III (S)	4.5.17 (S)			X
18	BATTERY GAS EMISSION TEST		X		3.8.10 (G), 3.8.12 (G)	4.5.18 (G)			
19	VIBRATION TEST		X		3.7 (G), 3.8.18 (N)	4.5.19 (S)			
20	HUMIDITY TEST		X		3.7 (G), 3.8.10 (G)	4.5.20 (G)			X

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TABLE I. Qualification inspection and conformance inspection of batteries – Continued.

Test number	Examinations and tests	Qualification inspection					Conformance inspection		
		Sample number			Requirement paragraph 1/	Test paragraph 1/	Group A sample 1 100%	Group B sample number	
		1	2	3				1	2
21	SALT FOG TEST	X			3.7 (G) 3.8.10 (G)	4.5.21 (G)		X	
22	GROUND STORAGE TEST			X 3/	3.8.13.1 (G)	4.5.22 (S)			
23	DEEP DISCHARGE RECOVERY TEST			X	3.7 (G), 3.8.13.3 (S)	4.5.27 (G)			
24	PHYSICAL INTEGRITY AT HIGH TEMPERATURE TEST	X			3.7 (G), 3.8.10 (G), Reqmt 1 of table III (S)	4.5.23 (S)			
25	SHOCK TEST (CRASH SAFETY)	X			3.8.10 (G) 3.8.15 (G)	4.5.25 (G)			
26	IMPACT RESISTANCE TEST (NONMETALLIC BATTERY CONTAINER)		4/		3.8.14 (G)	4.5.24 (G)			
27	WORMGEAR RETAINING PIN HARDNESS		5/		3.5.5 (S)	4.5.30 (G)			
28	SHELF LIFE TEST			X	3.8.9 (S), 3.8.13.2 (G)	4.5.26 (S)			
29	FINAL EXAMINATION	X	X	X	3.7 (G), 3.8.4 (G)	4.5.31 (G)		X	X

1/ The notations in parentheses specify the source of the paragraph. “(G)” means the paragraph as shown in MIL-PRF-8565. “(S)” shows that the paragraph has been modified by this specification sheet. “(N)” indicates that a new paragraph or requirement has been added by MIL-PRF-8565/15.

2/ See sample information in test paragraph.

3/ The manufacturer requesting qualification to this specification may submit an additional (fourth) test sample to expedite the qualification test program.

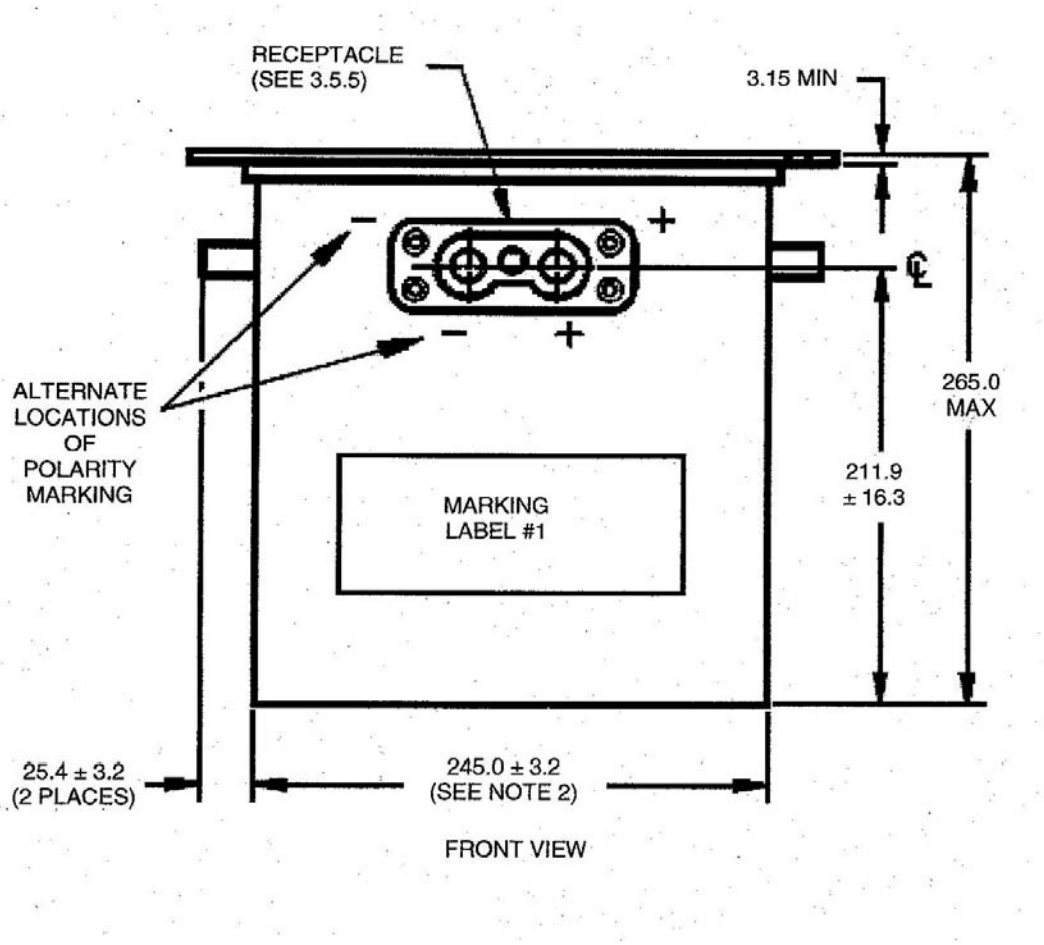
4/ For nonmetallic battery container designs, two outer nonmetallic battery containers shall be furnished for inspection.

5/ Five wormgear retaining pins shall be furnished for inspection.

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TABLE II. PIN D8565/15-1 random vibration test levels.

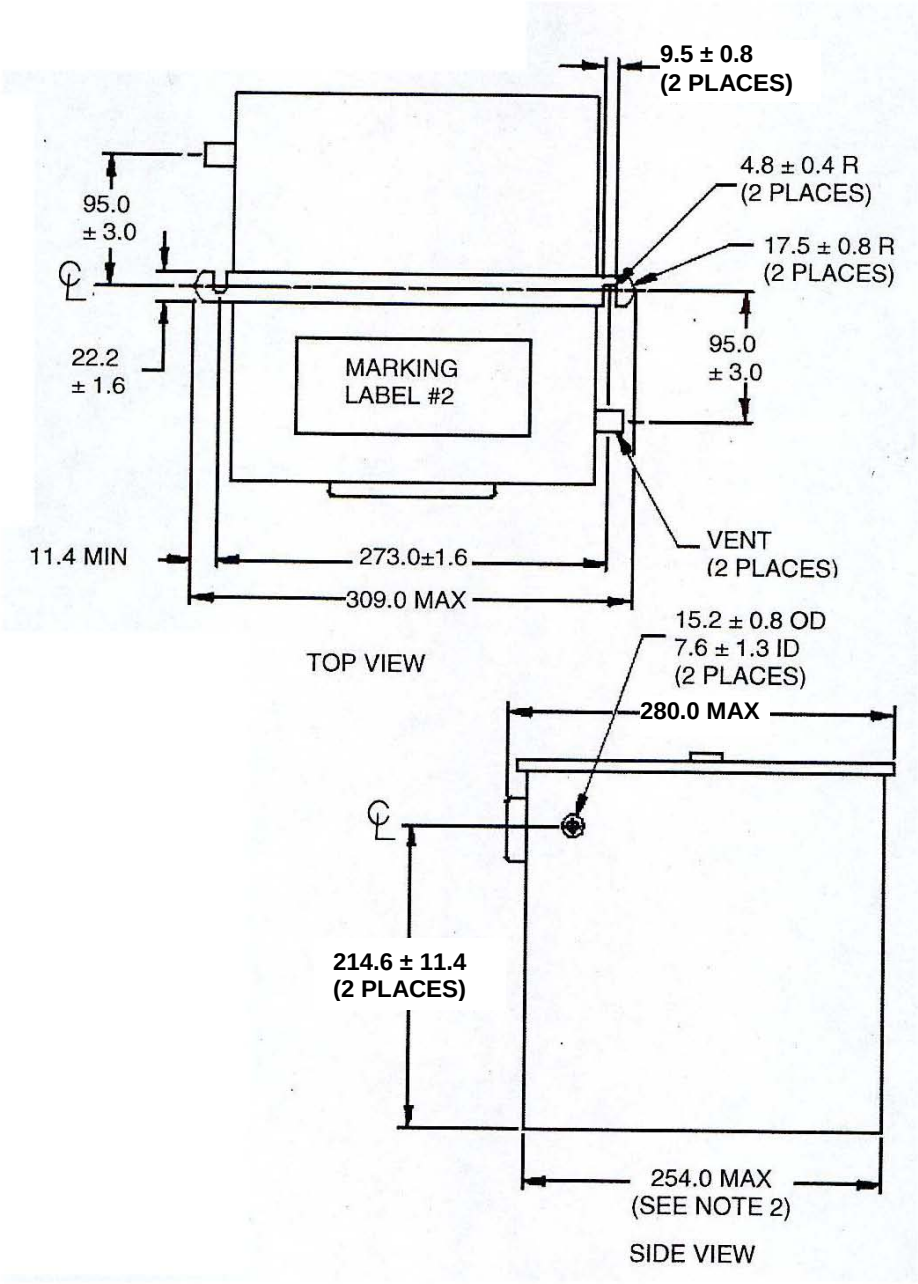
Requirement	Frequency 15 - 200 Hz	Frequency 200 - 300 Hz	Frequency 300 - 1000 Hz	Frequency 1000 - 2000 Hz	Overall 'G'rms
Test Level	0.04 g^2Hz	0.04 - 0.07 g^2Hz Slope + 4dB/Oct	0.07 g^2Hz	0.07 - 0.04 g^2Hz Slope - 6dB/Oct	9.84



NOTES:
1. All dimensions are in millimeters.
2. Allow additional 4 mm width for cover.
3. Abbreviations used in this figure are maximum (MAX), minimum (MIN) and part identification number (PIN).

FIGURE 1. PIN D8565/15-1 dimensions.

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NOTES:

1. All dimensions are in millimeters.
2. Allow additional 4 mm length for cover.
3. Abbreviations used in this figure are inside diameter (ID), outside diameter (OD), maximum (MAX), minimum (MIN) and radius (R).

FIGURE 1. PIN D8565/15-1 dimensions - Continued.

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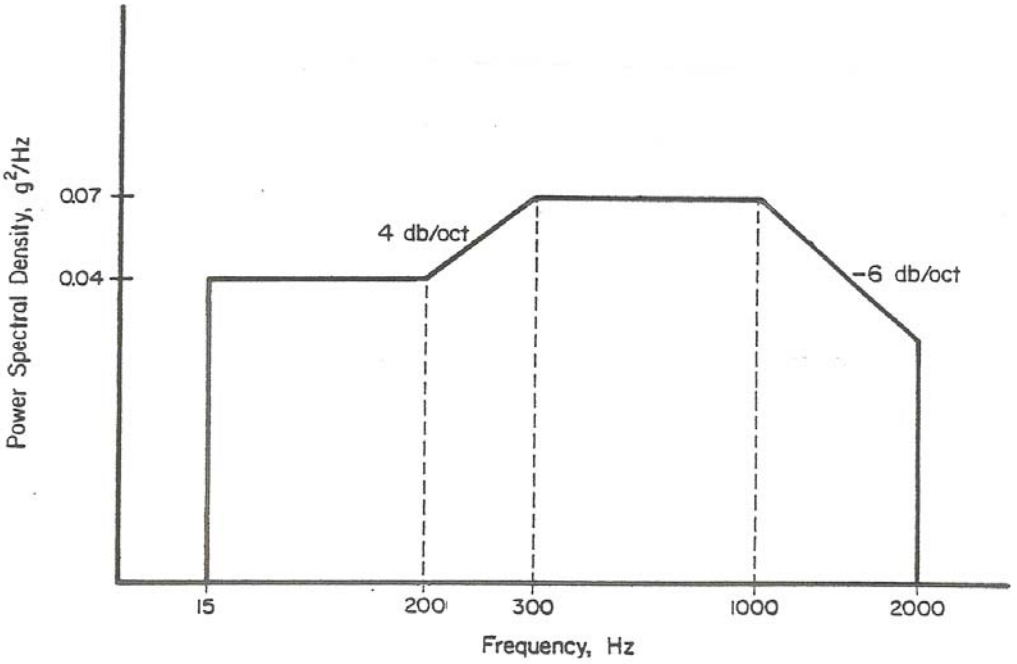


FIGURE 2. PIN D8565/15-1 random vibration curve.

6.19 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of marginal notations.

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